

University Interscholastic League
2024 – 2025 Junior High Number Sense Test C

Contestant's Number _____

Final		
2 nd		
1 st		
	Score	Initials

**Read Directions Carefully
Before Beginning Test**

**Do Not Unfold This Sheet
Until Told to Begin**

Directions: Do not turn this page until the person conducting this test gives the signal to begin. This is a ten-minute test. There are 80 problems. Solve accurately and quickly as many as you can in the order in which they appear. **ALL PROBLEMS ARE TO BE SOLVED MENTALLY.** Make no calculations with paper and pencil. Write only the answer in the space provided at the end of each problem. Problems marked with a (*) require approximate integral answers; any answer to a starred problem that is within five percent of the exact answer will be scored correct; all other problems require exact answers.

The person conducting this contest should explain these directions to the contestants.

Stop – Wait for Signal!

- | | |
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| <p>(1) $40 \div 0.25 =$ _____</p> <p>(2) $25 \times 0.24 =$ _____</p> <p>(3) $717 - 408 =$ _____</p> <p>(4) $639 \div 9 =$ _____</p> <p>(5) $\frac{3}{4} - \frac{7}{12} =$ _____</p> <p>(6) $5.75 - 0.75 =$ _____</p> <p>(7) $12 \times 16 =$ _____</p> <p>(8) $25^2 =$ _____</p> <p>(9) $16 + 18 \div 3 - 6 =$ _____</p> <p>*(10) $2025 \times 99 + 2024 =$ _____</p> <p>(11) $43 \times 47 =$ _____</p> <p>(12) The median of 10, 7, 24, 13 is _____</p> <p>(13) $105 \times \frac{7}{15} =$ _____</p> <p>(14) $44 \times 23 - 44 \times 18 =$ _____</p> <p>(15) $15 \times 10 \times 15 =$ _____</p> <p>(16) $318 + 58 + 42 =$ _____</p> <p>(17) $71^2 =$ _____</p> <p>(18) $34 + 28 + 22 + 16 + 10 =$ _____</p> <p>(19) MMXXV = _____ (Arabic Numeral)</p> | <p>*(20) $454 \times 329 + 34 =$ _____</p> <p>(21) $99 \times 96 =$ _____</p> <p>(22) If $5x - 13 = 32$, then $x^2 =$ _____</p> <p>(23) $5\frac{7}{12} + 6\frac{3}{4} =$ _____</p> <p>(24) An $8\frac{1}{3}\%$ sales tax on \$60 is \$ _____</p> <p>(25) $8\frac{1}{2} \times 2\frac{1}{2} =$ _____ (mixed number)</p> <p>(26) $1.25 \times 1.6 \times 10 =$ _____</p> <p>(27) 3 gallons = _____ cubic inches</p> <p>(28) The GCD of 12, 36 and 24 is _____</p> <p>(29) How many whole numbers evenly divide 44? _____</p> <p>*(30) $221 \times 636 + 44 =$ _____</p> <p>(31) $68 \times 48 =$ _____</p> <p>(32) If a square, with area 81, and equilateral triangle have the same perimeter, the triangle side = _____</p> <p>(33) If $1.75 - 0.25 = n$, the $n^{-1} =$ _____</p> <p>(34) $-13^2 =$ _____</p> <p>(35) 5 pints = _____ ounces</p> <p>(36) 2 inches = _____ centimeters</p> <p>(37) $58\frac{1}{3}\%$ of 48 is _____</p> |
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- (38) $876 \times 111 =$ _____
- (39) The area of a square with a diagonal of 14 is _____
- *(40) 5 miles = _____ feet
- (41) For the sequence: 2, **n**, 10, **q**, 26, 37, . . . **q + n** equals _____
- (42) $3^4 - 3^3 =$ _____
- (43) $41 \times 75 =$ _____
- (44) The number of elements in $\{2, 3, 5, 6\} \cap \{1, 2, 3, 6\}$ is _____
- (45) $2 \text{ (base 4)} \times 32 \text{ (base 4)} =$ _____ (base 4)
- (46) The length of the edge of a cube with surface area 600 is _____
- (47) The perimeter of a right triangle with leg 8 and hypotenuse 10 is _____
- (48) The smallest prime number larger than 80 is _____
- (49) The volume of a rectangular solid that measures 6 by 3 by x is 72. What is x ? _____
- *(50) $66500 \div 25 =$ _____
- (51) What is the area of the trapezoid with bases $4\frac{2}{3}$, $6\frac{2}{3}$ and height 9? _____
- (52) What is the 9th term in the sequence: 1, 2, 3, ...? _____
- (53) $101 \times 857 =$ _____
- (54) Three times a number added to 15 is 36. The number is _____
- (55) What is the perimeter of a rectangle with diagonal 10 and length 8? _____
- (56) The area of a rhombus with diagonals 12 and x is 48. What is x ? _____
- (57) $125 \times 32 =$ _____
- (58) $(7^4 + 7 \times 9) \div 4$ has a remainder of _____
- (59) If $28 - 2x > 44$, then $x <$ _____
- *(60) $8060 \div 79 \times 101 =$ _____
- (61) $32 + (-32) \div 8 =$ _____
- (62) $18\frac{4}{7}$ weeks = _____ days
- (63) 10 meters/sec = _____ km/hr
- (64) $0.1333\dots =$ _____ (common fraction)
- (65) $\frac{7}{5} + \frac{5}{7} =$ _____
- (66) The volume of a circular cylinder with radius 6 and length 25 is $k\pi$. What is k ? = _____
- (67) The sum of the whole number divisors of 12 is _____
- (68) $\frac{5! - 3!}{3!} =$ _____
- (69) 275% of 160 = _____
- *(70) 5 square yards = _____ inches
- (71) $286 \times 77 =$ _____
- (72) $8\frac{1}{3}\% =$ _____ (common fraction)
- (73) $45 \times 0.7333\dots =$ _____
- (74) The surface area of a sphere with a diameter of 12 is $k\pi$ and $k =$ _____
- (75) What is the probability of drawing a red ace from a standard deck of 52 cards? _____
- (76) If the shortest distance between the points (4, 3) and (0, **x**) is 5, what is **x**? _____
- (77) $4^3 \div 16 =$ _____
- (78) 14% of **n** equals 15% of 28. **n** = _____
- (79) $19 + 17 + 15 + \dots + 1 =$ _____
- *(80) $6875 \times 479 =$ _____

2024 – 2025 University Interscholastic League Junior High Number Sense Test C – Key

(1) 160	*(20) 141930 – 156870	(38) 97236	(59) -8
(2) 6	(21) 9504	(39) 98	*(60) 9790 – 10819
(3) 309	(22) 81	*(40) 25080 – 27720	(61) 28
(4) 71	(23) $12\frac{1}{3}; \frac{37}{3}$	(41) 22	(62) 130
(5) $\frac{1}{6}$	(24) 5.00	(42) 54	(63) 36
(6) 5	(25) $21\frac{1}{4}$	(43) 3075	(64) $\frac{2}{15}$
(7) 192	(26) 20	(44) 3	(65) $2\frac{4}{35}; \frac{74}{35}$
(8) 625	(27) 693	(45) 130	(66) 900
(9) 16	(28) 12	(46) 10	(67) 28
*(10) 192375 – 212623	(29) 6	(47) 24	(68) 19
(11) 2021	*(30) 133570 – 147630	(48) 83	(69) 440
(12) $11.5; 11\frac{1}{2}; \frac{23}{2}$	(31) 3264	(49) 4	*(70) 6156 – 6804
(13) 49	(32) 12	*(50) 2527 – 2793	(71) 22022
(14) 220	(33) $\frac{2}{3}$	(51) 51	(72) $\frac{1}{12}$
(15) 2250	(34) -169	(52) 9	(73) 33
(16) 418	(35) 80	(53) 86557	(74) 144
(17) 5041	(36) $5.08; 5\frac{2}{25}; \frac{127}{25}$	(54) 7	(75) $\frac{1}{26}$
(18) 110	(37) 28	(55) 28	(76) 0
(19) 2025		(56) 8	(77) 4
		(57) 4000	(78) 30
		(58) 0	(79) 100
			*(80) 3128469 – 3457781

Note: *(Number) x – y means an integer between x and y inclusive.

If an answer is of the type like $\frac{2}{3}$ it cannot be written as .666... or $\overline{.6}$.